The t_c-Cederbaum model: ghost Gutzwiller by direct minimization

(core fragment with $B_c = 1$, 3 and 5 quasiparticle orbitals)

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This note defines the Hamiltonian that is solved, the t_c-Cederbaum model (the Cederbaum model of core-hole satellites plus a weak core–virtual hopping t_c; $t_c = 0$ is the original Cederbaum model), its exact solution by diagonalization (ED), the ghost-Gutzwiller (gGut) variational problem used for it, the direct minimization that solves that problem, and the comparison of the gGut results with ED, for a core fragment without ghosts ($B_c = 1$) and with two and four ghosts ($B_c = 3$, 5). The aim of the test is to see whether gGut reproduces the satellite structure of the core-hole spectrum of this model. Quantities are per spin unless stated otherwise; solutions are real and paramagnetic.

1 The Hamiltonian

Three orbitals, c (core), o (occupied) and v (virtual), each with two spins. The t_c-Cederbaum model is $H_{\mathrm{tot}} = H + H_t$, where H is the Cederbaum model (the o–v hopping conditioned on a core hole) and H_t the added hopping:

$$\begin{aligned} H_{\mathrm{tot}} = {} & g \sum_{\sigma} (n_{v\sigma} - n_{o\sigma}) - E_c \sum_{\sigma} n_{c\sigma} - V \sum_{\sigma\sigma'} (1 - n_{c\sigma}) \left(c^{\dagger}_{v\sigma'} c_{o\sigma'} + \mathrm{h.c.} \right) \\ & + t_c \sum_{\sigma} \left(c^{\dagger}_{c\sigma} c_{v\sigma} + \mathrm{h.c.} \right), \end{aligned} \tag{1}$$

with $E_c > 0$, orbital gap g, core–valence coupling V, and a weak core–virtual hopping t_c. The prefactor $\sum_{\sigma}(1 - n_{c\sigma}) = 2 - n_c$ counts the core holes, so the o–v hopping acts only when the core is not full. The particle number and S_z are conserved. The core occupation n_c is conserved only at $t_c = 0$. Energies are measured in units of g ($g = 1$), so the parameters are the ratios E_c/g, V/g and t_c/g. All results below use $E_c/g = 10$, $V/g \in \{1/\sqrt{2}, \sqrt{2}\} = \{0.707, 1.414\}$ and $t_c/g \leq 0.5$; the ground state then lies in the sector with two electrons of each spin. The two values of V are chosen so that $\rho/\Omega = V/(\sqrt{2}\, g) = 0.5$ and 1.0, the ratios of the transition moment ρ to the excitation energy $\Omega = 2g$ used by Kockläuner and Golze for this model [2] (Sec. 6.3); E_c only shifts all poles rigidly at $t_c = 0$.

2 Exact diagonalization

Diagonalization. H_{tot} is diagonalized exactly in the 64-dimensional Fock space of six spin orbitals (Jordan–Wigner operators). At $t_c = 0$ the ground state is the product state $|220\rangle$ (notation $|n_c n_o n_v\rangle$: core and o doubly occupied, v empty) with $E = -2E_c - 2g$ and core occupation

$\nu_c = \langle n_c \rangle = 2$; for $t_c > 0$ it acquires a core-hole component and $\nu_c < 2$. The core-hole Green's function for spin $\uparrow$ is

$$G^h_{cc}(\omega) = \sum_n \frac{|\langle n|c_{c\uparrow}|\mathrm{GS}\rangle|^2}{\omega + (E_n - E_{\mathrm{GS}}) - i\eta}, \tag{2}$$

with poles at $\omega = -E_i$ and weights p_i. At $t_c = 0$ there are three poles; with $\alpha = V/g$,

$$\begin{aligned} E_1 &= E_c + 2g - 2g\sqrt{1+\alpha^2}, & p_1 &= \frac{(1+\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \\ E_2 &= E_c + 2g, & p_2 &= \frac{\alpha^2}{2(1+\alpha^2)}, \\ E_3 &= E_c + 2g + 2g\sqrt{1+\alpha^2}, & p_3 &= \frac{(1-\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \end{aligned} \tag{3}$$

which the ED reproduces to 9×10^{-15}. For $t_c > 0$ the poles move at $O(t_c^2)$ and the middle pole splits into two lines $\sim 10^{-4}$ apart (Tab. 2); Tab. 1 gives the ground-state quantities.

V/g	t_c/g	$(E_0 + 2E_c + 2g)/g$	$2 - \nu_c$	$\rho^{cv} = \langle c^\dagger_{c\uparrow} c_{v\uparrow}\rangle$
0.707	0.5	-4.55×10^{-2}	4.15×10^{-3}	-4.54×10^{-2}
0.707	0.2	-7.30×10^{-3}	6.67×10^{-4}	-1.82×10^{-2}
0.707	0.1	-1.82×10^{-3}	1.67×10^{-4}	-9.12×10^{-3}
0.707	0.05	-4.56×10^{-4}	4.17×10^{-5}	-4.56×10^{-3}
1.414	0.5	-4.60×10^{-2}	4.27×10^{-3}	-4.59×10^{-2}
1.414	0.2	-7.37×10^{-3}	6.87×10^{-4}	-1.84×10^{-2}
1.414	0.1	-1.84×10^{-3}	1.72×10^{-4}	-9.22×10^{-3}
1.414	0.05	-4.61×10^{-4}	4.30×10^{-5}	-4.61×10^{-3}

Table 1: Exact ground state of H_{tot}: $2 - \nu_c \propto t_c^2$ and $\rho^{cv} \propto t_c$.

V/g		$-E_1$	$-E_2$ (split pair)	$-E_3$	p_1
0.707	$t_c \to 0$ (Eq. 3)	−9.5505	−12	−14.4495	0.8249
	$t_c/g = 0.5$	−9.5777	−12.0441, −12.0477	−14.5121	0.8268
	$t_c/g = 0.1$	−9.5516	−12.0018, −12.0019	−14.4520	0.8250
	$t_c/g = 0.05$	−9.5508	−12.0004, −12.0005	−14.4501	0.8249
1.414	$t_c \to 0$ (Eq. 3)	−8.5359	−12	−15.4641	0.6220
	$t_c/g = 0.5$	−8.5707	−12.0406, −12.0510	−15.5198	0.6263
	$t_c/g = 0.1$	−8.5373	−12.0016, −12.0020	−15.4663	0.6222
	$t_c/g = 0.05$	−8.5362	−12.0004, −12.0005	−15.4647	0.6221

Table 2: Exact hole poles of G^h_{cc} at $V/g = 0.707$ and 1.414 (weights $> 10^{-3}$); the split pair has total weight $p_2 = \alpha^2/[2(1+\alpha^2)] = 0.1667$ and 0.3333, respectively.

3 The ghost-Gutzwiller variational problem

Fragments and quasiparticle space. Each orbital $I \in \{c, o, v\}$ is a fragment with B_I quasiparticle orbitals $d_{Ia\sigma}$, $a = 1, \ldots, B_I$: $B_o = B_v = 1$ (no local interaction) and $B_c \in \{1, 3, 5\}$, i.e. $B_c - 1$ ghost orbitals on the core. The Gutzwiller wave function is $|\Psi_G\rangle = \prod_I P_I |\psi_{qp}\rangle$ with $|\psi_{qp}\rangle$

a Slater determinant that fills $N_{\rm occ} = 2 + (B_c - 1)/2$ of the $B_c + 2$ levels per spin. Its one-body density matrix is $\rho^{IJ}_{ab} = \langle d^\dagger_{Ia\sigma} d_{Jb\sigma}\rangle$.

Embedding states. The projector of each fragment is represented by a real normalized state Φ_I in the Fock space of the physical orbital I plus B_I bath orbitals per spin, restricted to the half-filled sector with $(1 + B_I)/2$ electrons of each spin (B_I odd). Writing $c_{I\sigma}$ for the physical and $d_{Ia\sigma}$ for the bath orbitals, the following quantities are expectation values in Φ_I (spin averaged):

$$\begin{aligned} D_{I,a} &= \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} c_{I\sigma}\rangle, \quad \Delta^{bb}_{I,ab} = \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} d_{Ib\sigma}\rangle, \\ \nu_c &= \langle n_c\rangle, \qquad\qquad\qquad K_{\gamma\delta} = \Big\langle n_c \sum_\sigma d_{c\delta\sigma} d^\dagger_{c\gamma\sigma}\Big\rangle, \end{aligned} \tag{4}$$

and $E^{loc}_I = \langle H^{loc}_I\rangle$ with $H^{loc}_c = -E_c n_c$, $H^{loc}_o = -g n_o$, $H^{loc}_v = +g n_v$ ($n_I = \sum_\sigma n_{I\sigma}$). From them:

$${}^I\Delta = \mathbb{I} - \Delta^{bb}_I, \qquad S_I = \big[{}^I\Delta(\mathbb{I} - {}^I\Delta) + \varepsilon\,\mathbb{I}\big]^{1/2}, \qquad R_I = S_I^{-1} D_I, \qquad C = (\mathbb{I} - {}^c\Delta + \varepsilon\,\mathbb{I})^{-1}, \tag{5}$$

where ε is a small regularizing shift (10^{-9} unless stated) and $R_I \in \mathbb{R}^{B_I}$ is the renormalization vector, $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$. The non-local core–valence correlation enters through the tensor

$$\bar\Theta = \tfrac{1}{2} S_c^{-1} K S_c^{-1} - \nu_c\, C, \qquad u = \rho^{cv} R_v, \quad w = \rho^{co} R_o, \tag{6}$$

which follows from evaluating $\langle\Psi_G| n_{c\sigma} c^\dagger_{o\sigma'} c_{v\sigma'} |\Psi_G\rangle$ with the Gutzwiller approximation (derived in `main.tex`, Eqs. ThetaEff–ThetaEqn, and `SI.tex`, S2).

Energy functional. The energy is $E = \langle\Psi_G|H_{\rm tot}|\Psi_G\rangle$, evaluated term by term from Eq. (1) with $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$ and Wick's theorem on $|\psi_{qp}\rangle$.

- *Local terms.* $g(n_v - n_o) - E_c n_c = \sum_I H^{loc}_I$ is evaluated in the embedding states, $\sum_I E^{loc}_I$.
- *Core–virtual hopping.* Per spin, $\langle c^\dagger_{c\sigma} c_{v\sigma}\rangle = R_c^T \rho^{cv} R_v$. The sum over the two spins and the hermitian conjugate give a factor 4: $4t_c\, R_c^T \rho^{cv} R_v$.
- *Conditioned o–v hopping.* The weight $(2 - n_c)$ multiplies $\sum_{\sigma'}(c^\dagger_{v\sigma'} c_{o\sigma'} + \text{h.c.})$ and the expectation value does not factorize, since n_c fluctuates and is correlated with the hopping. Per spin, $\langle (2 - n_c)\, c^\dagger_{v\sigma} c_{o\sigma}\rangle = (2 - \nu_c)\, R_v^T \rho^{vo} R_o + w^T \bar\Theta\, u$: the first part is the mean-field one, with the average number of core holes $2 - \nu_c$; the second is the connected core–valence correlation, built from $\bar\Theta$, u and w defined above. Spin sum and hermitian conjugate again give a factor 4.

Adding the three contributions with the prefactors of Eq. (1):

$$E = \underbrace{\sum_I E^{loc}_I}_{\text{local}} + \underbrace{4t_c\, R_c^T \rho^{cv} R_v}_{t_c\ \text{hopping}} - \underbrace{4V(2 - \nu_c)\, R_v^T \rho^{vo} R_o}_{o\text{–}v\ \text{mean field}} - \underbrace{4V\, w^T \bar\Theta\, u}_{\text{connected}}. \tag{7}$$

Written out with the definitions above, the dependences are: E^{loc}_I, ν_c, Δ^{bb}_I and K are expectation values in Φ_I; $R_I = S_I^{-1} D_I$ with S_I a function of Δ^{bb}_I; $\bar\Theta$ a function of ν_c, K and ${}^c\Delta$; $u = \rho^{cv} R_v$ and $w = \rho^{co} R_o$. So E depends on the embedding states only through E^{loc}_I, D_I, Δ^{bb}_I, ν_c and K, and on $|\psi_{qp}\rangle$ only through ρ.

The limit $t_c = 0$. At $t_c = 0$ the core occupation is conserved and the exact ground state is the product state $|220\rangle$ of Sec. 2, with $\nu_c = 2$. There is no core–valence correlation, so the core is coherent neither with v nor with o: $\rho^{cv} = \rho^{co} = 0$ and $u = w = 0$. Term by term in Eq. (7):

- the t_c hopping vanishes through its explicit factor t_c;
- the mean-field o–v term vanishes through $2 - \nu_c = 0$;
- the connected term vanishes through $u = w = 0$.

Hence

$$E\big|_{t_c=0} = \sum_I E_I^{loc} = -2E_c - 2g, \tag{8}$$

which is the exact energy. Two consequences follow. First, V does not appear: every V term of Eq. (7) is zero. Second, $R_c = S_c^{-1} D_c$ appears in Eq. (7) *only* in the t_c hopping term (the other quantities, E_I^{loc}, ν_c, Δ_I^{bb}, K, do not involve D_c, and u, w contain R_v, R_o but not R_c), so with that term gone E is independent of D_c and the minimization leaves R_c undetermined. At $V = 0$ and $t_c \neq 0$ the situation is different: the two V terms drop out, but $E = \sum_I E_I^{loc} + 4t_c\, R_c^T \rho^{cv} R_v$ still depends on R_c, which is therefore determined by the hopping.

These two consequences are the reason for the hopping t_c. The quantity of interest is not the ground-state energy but the core-hole Green's function G^h_{cc}, whose three poles are final-state physics: the three $N - 1$ eigenstates reached when a core electron is removed and V then moves electrons between o and v. gGut does not compute $N - 1$ states; the spectrum comes from the quasiparticle Hamiltonian H_{qp} (Sec. 4), whose parameters R_I, λ_I are fixed by the ground-state variational problem. At $t_c = 0$ that problem contains no V and leaves R_c free, so the G^h_{cc} built from R_c carries no information about the satellites, although the ground state itself is exact. A weak t_c cures both defects at once: it makes R_c determined and slightly correlates the ground state ($\nu_c < 2$), so that the V terms of Eq. (7) switch on and R_c, H_{qp} acquire V-dependent values. It is a probe, not a change of the question: the exact poles move only at $O(t_c^2)$ (Sec. 2), so a method that captures the satellites should reproduce them at small t_c.

The hopping has to couple $|220\rangle$ to other states, which fixes its form. A hopping $t_c(c_c^\dagger c_o + \text{h.c.})$ is useless: both orbitals are full in $|220\rangle$, so it annihilates the state, and $|220\rangle$ remains an exact eigenstate for every t_c (checked by ED). With $c_v^\dagger c_c$, instead, a core electron moves to the empty v at an energy cost $E_c + g$, so the ground state acquires a core-hole component of amplitude $\sim t_c/(E_c + g)$ and $2 - \nu_c = O(t_c^2)$. This is exactly the combination that Eq. (7) needs to become V-dependent: the term $4t_c\, R_c^T \rho^{cv} R_v$ determines R_c, and $2 - \nu_c \neq 0$ activates the o–v terms.

Variational problem.

$$\min_{\{\Phi_I\},\ \psi_{qp}} E \qquad \text{subject to} \qquad \langle \Phi_I | \Phi_I \rangle = 1, \qquad \rho^{II} = {}^I\Delta(\Phi_I) \quad (I = c, o, v). \tag{9}$$

The second constraint ties the local quasiparticle density matrix to the embedding state; it has $B_I(B_I+1)/2$ independent components per fragment (8 for $B_c = 3$). Multipliers λ_I of this constraint define the quasiparticle Hamiltonian H_{qp} of Sec. 4.

4 Direct minimization

Problem (9) is solved directly (`not_imp_gga_cederbaum.py`), as in the direct-minimization solver used for the Hubbard dimer, without any self-consistency cycle.

Variables. The embedding states are parametrized by unnormalized real vectors x_I in the half-filled sector ($\Phi_I = x_I/|x_I|$; dimensions 36, 4, 4 for $B_c = 3$ and 400, 4, 4 for $B_c = 5$), and the Slater determinant by a real $(B_c + 2) \times N_{\mathrm{occ}}$ matrix X with $\rho = X(X^TX)^{-1}X^T$. The constraints $\rho^{II} = {}^I\Delta(\Phi_I)$ are imposed as equalities (SLSQP, tolerance 10^{-13} on the energy). Everything is evaluated with sector-projected dense matrices of the operators defined in Sec. 3.

Gradients. The analytic gradient of Eq. (7) follows from the chain rule through $R_I(D_I, \Delta_I^{bb})$, $\bar{\Theta}(\nu_c, K, {}^c\Delta)$ and $\rho(X)$:

$$dE = \sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\Lambda_I\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc} + 2\,\mathrm{Tr}[H_{\mathrm{off}}\, d\rho], \tag{10}$$

with $h_I = S_I^{-1}X_I$ and $X_I = \frac{1}{4}\,\partial E/\partial R_I$ given by

$$\begin{aligned} X_c &= t_c\,\rho^{cv}R_v, \qquad X_o = -V\big[(2-\nu_c)\rho^{ov}R_v + \rho^{oc}\bar{\Theta}u\big], \\ X_v &= t_c\,\rho^{vc}R_c - V\big[(2-\nu_c)\rho^{vo}R_o + \rho^{vc}\bar{\Theta}w\big], \end{aligned} \tag{11}$$

The matrices $\Lambda_I = -\frac{1}{2}\nabla^s_{{}^I\Delta}E$ (derivative of E with respect to ${}^I\Delta$ at fixed D_I, through R_I, S_I and $\bar{\Theta}$) are

$$\begin{aligned} \Lambda_{o,v} &= G\big({}^I\Delta,\, h_I R_I^T\big), \\ \Lambda_c &= G\big({}^c\Delta,\, h_c R_c^T - \tfrac{1}{4}Y\big) - V\nu_c\, C\,(uw^T + wu^T)\,C, \\ Y &= V\big(S_c^{-1}K\,\Pi + \Pi\,K\,S_c^{-1}\big), \quad \Pi = xy^T + yx^T, \quad x = S_c^{-1}w,\ y = S_c^{-1}u, \end{aligned} \tag{12}$$

where $G(\Delta, M) = W\,[(\tilde{M} + \tilde{M}^T) \circ F]\,W^T$ is twice the symmetrized gradient of $\mathrm{Tr}[M\,S(\Delta)]$, $\Delta = W\,\mathrm{diag}(n_a)W^T$, $\tilde{M} = W^TMW$, $F_{ab} = (1 - n_a - n_b)/(s_a + s_b)$, $s_a = \sqrt{n_a(1-n_a) + \varepsilon}$ and $\circ$ is the elementwise product. The remaining derivatives are

$$\beta = \frac{\partial E}{\partial \nu_c} = 4V\big(R_v^T\rho^{vo}R_o + w^TCu\big), \qquad \Xi = \nabla_K E = -V\,\Pi, \tag{13}$$

and H_{off} has the off-diagonal blocks $H^{co} = -V(\bar{\Theta}u)R_o^T$, $H^{cv} = t_cR_cR_v^T - V(\bar{\Theta}w)R_v^T$, $H^{ov} = -V(2-\nu_c)R_oR_v^T$ (and their transposes), i.e. $\frac{1}{4}\partial E/\partial\rho^{IJ} = H^{IJ}$. The Jacobian of the constraints is also analytic. Gradients and Jacobian are verified against finite differences in Sec. 5.

Lagrangian and its differential. With the multipliers λ_I (symmetric $B_I \times B_I$, one per independent component of the constraint $\rho^{II} = {}^I\Delta(\Phi_I)$) the Lagrangian of problem (9) is

$$\mathcal{L} = E\big(\{\Phi_I\}, \rho\big) + 2\sum_I \mathrm{Tr}\big[\lambda_I\big({}^I\Delta(\Phi_I) - \rho^{II}\big)\big], \tag{14}$$

where the normalization of Φ_I and the idempotency of ρ carry no multipliers: they are built into the variables, $\Phi_I = x_I/|x_I|$ and $\rho = X(X^TX)^{-1}X^T$. Since ${}^I\Delta = \mathbb{I} - \Delta_I^{bb}$, the constraint term has the differential $-2\mathrm{Tr}[\lambda_I(d\rho^{II} + d\Delta_I^{bb})]$. Adding it to Eq. (10), the terms in $d\Delta_I^{bb}$ combine into $2\mathrm{Tr}[(\Lambda_I - \lambda_I)\,d\Delta_I^{bb}]$ and the terms in $d\rho$ into $2\mathrm{Tr}[(H_{\mathrm{off}} - \mathrm{blockdiag}(\lambda_I))\,d\rho]$:

$$d\mathcal{L} = \underbrace{\sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\lambda_I^c\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc}}_{\text{variation of the } \Phi_I} + \underbrace{2\,\mathrm{Tr}[H_{qp}\, d\rho]}_{\text{variation of } X}, \tag{15}$$

with the two combinations

$$\lambda_I^c = \Lambda_I - \lambda_I, \qquad H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I). \tag{16}$$

These are the bath potential and the quasiparticle Hamiltonian of the self-consistency loop; here they appear as the coefficients of $d\mathcal{L}$. Stationarity of $\mathcal{L}$ is $d\mathcal{L} = 0$ for every variation of the Φ_I and of ρ within the projectors:

- *Variation of X.* $2\text{Tr}[H_{qp}\, d\rho] = 0$ for all variations of the projector, i.e. $[H_{qp}, \rho] = 0$: ρ is built from eigenvectors of its own H_{qp}.

- *Variation of Φ_I.* D_I, Δ_I^{bb}, ν_c, K and E_I^{loc} are expectation values in Φ_I (Sec. 3), so the first group of Eq. (15) is the differential of $\langle\Phi_I|H_I^{imp}|\Phi_I\rangle$ at fixed coefficients, i.e. $2\langle d\Phi_I|H_I^{imp}|\Phi_I\rangle$, with

$$\begin{aligned} H_I^{imp} = H_I^{loc} &+ \sum_a h_{I,a} \sum_\sigma \left(d_{Ia\sigma}^\dagger c_{I\sigma} + c_{I\sigma}^\dagger d_{Ia\sigma}\right) + \sum_{ab} \lambda_{I,ab}^c \sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma} \\ &+ \delta_{Ic}\Big[(\alpha + 2\text{Tr}\Xi)\, n_c - \sum_{\gamma\delta} \Xi_{\gamma\delta}\, n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\Big], \end{aligned} \tag{17}$$

  where $n_c = \sum_\sigma c_{c\sigma}^\dagger c_{c\sigma}$ and $H_c^{loc} = -E_c n_c$, $H_o^{loc} = -g\, n_o$, $H_v^{loc} = +g\, n_v$. The coefficients are

$$h_I = S_I^{-1} X_I, \qquad \lambda_I^c = \Lambda_I - \lambda_I, \qquad \alpha = \beta = 4V\left(R_v^T \rho^{vo} R_o + w^T C u\right), \qquad \Xi = -V\,\Pi, \tag{18}$$

  with X_I, Λ_I, Π, β as in Eq. (10) and below. Each term is the coefficient of one expectation value in the first group of Eq. (15): $h_{I,a}$ multiplies $\langle\sum_\sigma (d_{Ia\sigma}^\dagger c_{I\sigma} + \text{h.c.})\rangle = 4D_{I,a}$; $\lambda_{I,ab}^c$ multiplies $\langle\sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma}\rangle = 2\Delta_{I,ab}^{bb}$; $\alpha + 2\text{Tr}\Xi$ multiplies ν_c; and $-\Xi_{\gamma\delta}$ multiplies $\langle n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\rangle = 2\nu_c\delta_{\gamma\delta} - K_{\gamma\delta}$, which with the $2\text{Tr}\Xi\,\nu_c$ part reproduces $\text{Tr}[\Xi\, dK]$. A constant from normal ordering, $-2\text{Tr}\lambda_I^c$, is dropped. Its vanishing for all $d\Phi_I \perp \Phi_I$ means that Φ_I is an eigenstate of H_I^{imp}. **Note on the excited state of Φ_c.** Φ_c is not the lowest eigenstate: for $B_c = 3$, $t_c/g = 0.5$ the direct-minimization Φ_c is the fourth ($V/g = 0.707$) and the sixth ($V/g = 1.414$) of the 36 eigenstates of H_c^{imp} in energy order, with the lowest eigenvalue 2.3 and 8.1 below it, whereas Φ_o and Φ_v are ground states (`excited_phi_cederbaum.tex`, `excited_check_golze.py`). The impurity loop, which takes the lowest state, cannot follow it.

The self-consistency loop enforces these conditions one after the other (diagonalizing H_{qp}, solving for Φ_I); the direct minimization does not diagonalize either. It passes E, dE (Eq. (10)) and the constraints $\kappa = \rho^{II} - {}^I\Delta(\Phi_I)$ with their Jacobian ($d\kappa = d\rho^{II} + d\Delta_I^{bb}$) to SLSQP, which drives the gradient to a constrained stationary point. The multipliers are recovered afterwards from the first group of Eq. (15): stationarity in Φ_I of $E + \sum_k \mu_k\kappa_k$ is solved for μ in the least-squares sense (below). The eigenstate Φ_I is not required to be the lowest one of H_I^{imp}.

Note. In standard gGA, H_I^{imp} contains only H_I^{loc}, the hybridization h_I and the bath term λ_I^c, and the other fragments enter through h_I alone (the t_c hopping does this). The conditioned o–v hopping multiplies a bilinear by n_c, so its average also depends on ν_c and K (through $\bar{\Theta}$, from `main.tex`, `SI.tex`). This adds the n_c and $n_c d^\dagger d$ terms of H_c^{imp} and makes its coefficients α, Ξ, λ_c^c depend on o and v (through u, w, ρ^{vo}), and h_o, h_v on the core.

Starting points. The problem is non-convex, so several random starts are run per parameter set (embedding states from random small Hamiltonians, X from a random symmetric matrix); a point is accepted if the constraint residual is $< 10^{-9}$, and the lowest accepted energy is kept. Seeding from the previous t_c can end in a poor local minimum (Sec. 6.1), so points are judged by their residuals and not by the energy alone.

$B_c = 5$**.** Random starts rarely converge at $B_c = 5$ (436 variables), so the search is seeded from a $B_c = 3$ solution with the two new ghosts asleep: $\Phi_c^{(5)} = \Phi_c^{(3)} \otimes |\text{ghost}\rangle$ (one extra bath orbital filled, one empty) and $\rho^{(5)} = \rho^{(3)} \oplus \text{diag}(0, 1)$. This point is feasible and has the $B_c = 3$ energy; it is perturbed randomly and minimized. Sparse operators and caching make a $B_c = 5$ run take about 2 s.

Quasiparticle Hamiltonian and spectrum. The multipliers are recovered from the stationarity of $E + \sum_k \mu_k \kappa_k$ in Φ (least squares over the constraint components $\kappa_k = \rho^{II}_{ab} - {}^I\Delta_{ab}$; the residual of this fit is reported as the KKT residual), with $\lambda_{aa} = -\mu_{aa}/2$, $\lambda_{a<b} = -\mu_{ab}/4$. The quasiparticle Hamiltonian is $H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I)$, whose commutator with ρ vanishes at a solution (reported as $\|[H_{qp}, \rho]\|$). The hole Green's function of the core orbital is

$$G^h_{cc}(\omega) = \sum_{n \le N_{\text{occ}}} \frac{|R_c^T \phi_n^{(c)}|^2}{\omega - \epsilon_n - i\eta}, \tag{19}$$

with ϵ_n, ϕ_n the eigenvalues (ascending) and eigenvectors of H_{qp} and $\phi_n^{(c)}$ the core block of ϕ_n; the poles ϵ_n are compared with $-E_i$ and the weights with p_i [Eq. (3)]. The total weight of G_{cc} is $R_c^T R_c$. The constraint Jacobian is rank deficient at solutions with pure ghost occupations (eigenvalues 0 or 1 of ${}^c\Delta$): some multiplier components are then undetermined (Sec. 6.2).

5 Checks

The analytic gradients of Sec. 4 agree with central finite differences of Eq. (7) at random points, for $B_c = 1, 3, 5$, to $\le 4 \times 10^{-10}$ for each identity (H^{IJ}, $S_I h_I$, $G(\Delta, M)$, the ${}^c\Delta$ gradient at fixed R_c, $\beta = \partial E/\partial \nu_c$, $\Xi = \nabla_K E$), and the complete gradient of the minimization to a relative error of 4×10^{-7} (limited by the stiffness of S^{-1} for nearly pure ${}^I\Delta$); the constraint Jacobian to 3×10^{-10}. The limits are reproduced: at $V = 0$, $E - E_{\text{ref}} = +4.4 \times 10^{-8}$ for $B_c = 1$ and 3 (the effect of $\varepsilon = 10^{-9}$; $\nu_c = 1.995893$ and 1.999835 at $t_c/g = 0.5$ and 0.1); at $t_c = 0$ the energy is exact ($< 10^{-15}$ at $B_c = 1$; 10^{-13} and 10^{-11} at $B_c = 3$ for $V/g = 0.707$ and 1.414) and $\nu_c = 2$. At $t_c = 0$, R_c is not determined by the stationarity conditions and V does not enter (Sec. 3). The accepted $B_c = 3$ and 5 points have constraint residual $\lesssim 10^{-9}$ (typically 10^{-14}) and KKT residual $\le 2.5 \times 10^{-6}$.

6 Results for $B_c = 1$, 3 and 5

$B_c = 3$ and 5 mean two and four ghost orbitals on the core, $B_c = 1$ none (standard Gutzwiller). $B_c = 3$ and 5 are from the direct minimization ($B_c = 5$ seeded from the $B_c = 3$ solution, plus random starts); $B_c = 1$ from the converged loop, which the direct minimization reproduces to $\le 1.5 \times 10^{-14}$ in E at $t_c/g = 0.1$. All solutions used in the figures and tables are saved (`cederbaum_data/ghost_solutions.npz`).

6.1 Energy and ν_c

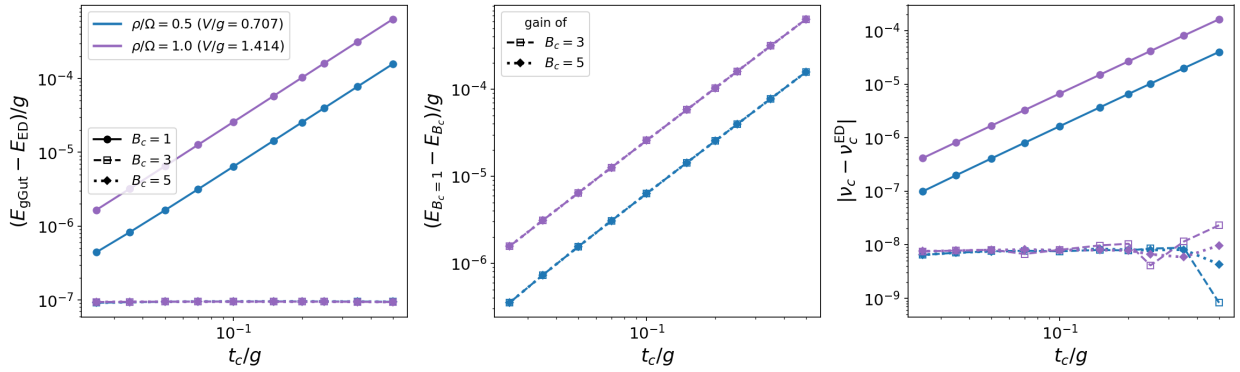


Figure 1: Left: $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ for $B_c = 1$ (full lines), $B_c = 3$ (dashed, open squares) and $B_c = 5$ (dotted, diamonds); blue: $\rho/\Omega = 0.5$ ($V/g = 0.707$), purple: $\rho/\Omega = 1.0$ ($V/g = 1.414$). Middle: energy gained by the ghosts, $E_{B_c=1} - E_{B_c}$ (the two coincide). Right: $|\nu_c - \nu_c^{\mathrm{ED}}|$. Log–log axes; $E_c/g = 10$.

- $B_c = 1$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} \simeq 1.3 \times 10^{-3}\, V^2 t_c^2$ over $t_c/g \in [0.025, 0.5]$ (exponent 1.97 and 1.99), and $|\nu_c - \nu_c^{\mathrm{ED}}| \propto t_c^2$.
- $B_c = 3$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} = 9.2$–$9.6 \times 10^{-8}$ for all (V, t_c), flat; $|\nu_c - \nu_c^{\mathrm{ED}}| \le 2.3 \times 10^{-8}$. This floor is the regularization shift, not the ansatz.
- $B_c = 5$**:** the same energy at all 20 points ($E_{B_c=5} - E_{B_c=3}$ between -6×10^{-11} and $+6 \times 10^{-13}$; KKT $\le 2.3 \times 10^{-6}$); 5–9 of 14 runs per point converged to the same energy, and 3 of 120 random starts. The extra ghosts do not lower the energy.

The ghosts thus recover 79–99.99% of the $B_c = 1$ error (the smallest fractions at the smallest t_c, where the shift floor is comparable to the error). Poor local minima from continuation ($(V/g, t_c/g) = (0.707, 0.025)$, $(1.414, 0.07)$) were recomputed from random starts; one $B_c = 1$ loop point needed mixing 0.3; two $B_c = 5$ seeds were obtained by continuation.

6.2 Hole spectral function

		ED main pole		gGut $B_c = 1$		gGut $B_c = 3$		gGut $B_c = 5$	
V/g	t_c/g	position	weight	position	weight	position	weight	position	weight
0.707	0.1	−9.5516	0.8250	−10.0007	0.9999	−9.9355	0.9940	−9.9355	0.9940
0.707	0.05	−9.5508	0.8249	−9.9997	0.9999	−9.9330	0.9939	−9.9330	0.9939
1.414	0.1	−8.5373	0.6222	−10.0005	0.9999	−9.7458	0.9767	−9.7458	0.9767
1.414	0.05	−8.5362	0.6221	−9.9997	0.9999	−9.7435	0.9766	−9.7434	0.9766

Table 3: Dominant hole pole of G_{cc}. ED additionally has satellites at about −12.0 (total weight 0.167 and 0.333 for $V/g = 0.707$, 1.414) and at −14.45 and −15.47 (weights 0.008 and 0.045); gGut has none: all other occupied levels have weight $\le 1.5 \times 10^{-6}$ ($B_c = 1$) and $\le 2.2 \times 10^{-6}$ ($B_c = 3$, 5).

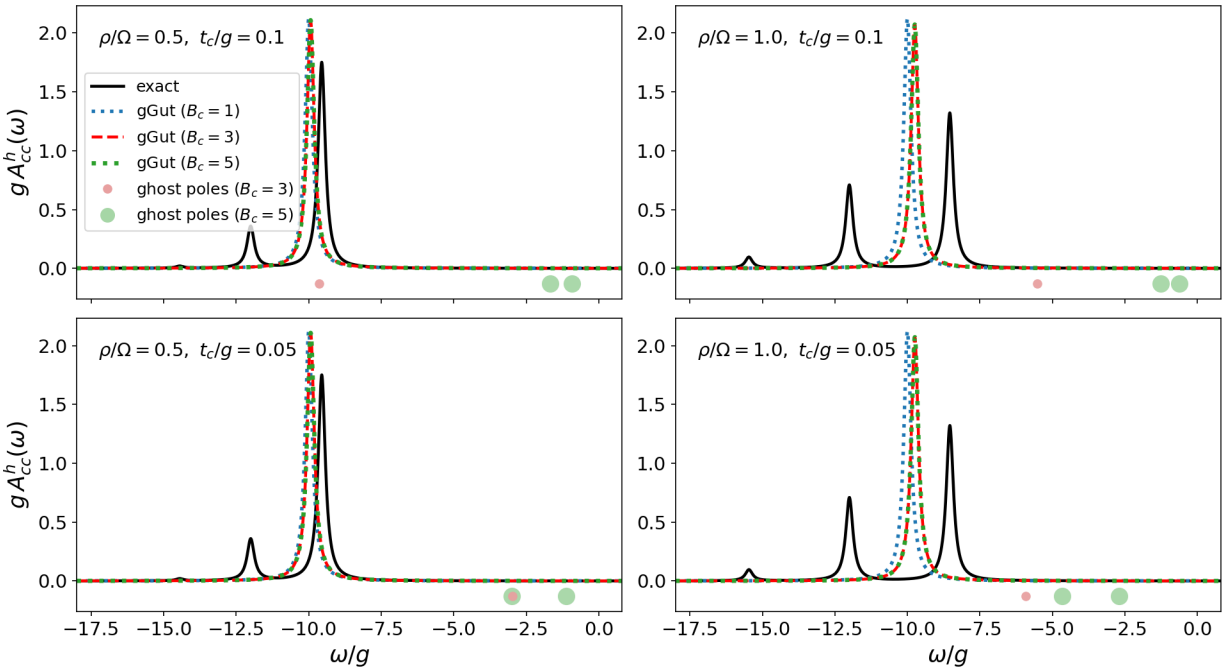


Figure 2: Hole spectral function $A^h_{cc}(\omega) = \sum_n w_n\,(\eta/\pi)/[(\omega-\omega_n)^2+\eta^2]$, $\eta = 0.15$: exact (solid), gGut $B_c = 1$ (blue dotted), $B_c = 3$ (red dashed), $B_c = 5$ (green dotted, on top of $B_c = 3$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$ ($V/g = 0.707$), 1.0 ($V/g = 1.414$); $E_c/g = 10$. Dots below zero: the occupied muted poles (zero-weight ghost levels of H_{qp}, defined in the text), as returned by the minimizer: one at $B_c = 3$ (small, rose), two at $B_c = 5$ (large, green); $B_c = 1$ has none. Their positions are not fixed by the solution (see text) and they carry no weight.

Result. The $B_c = 1$ pole sits at -10.000 for every V: the bare core level. At $B_c = 3$ and 5 (identical to 10^{-4}) the single pole moves with V towards the exact main pole but covers only 15–18% of the exact displacement from $-E_c$ (0.067 of 0.449 and 0.257 of 1.464 at $t_c/g = 0.05$), and its weight (0.994, 0.977) stays far above the exact one (0.825, 0.622). *No satellites appear*; the ghosts only add V dependence to the main pole.

Why there is no weight on the ghosts (interpretation). The weights in Eq. (19) are $|R_c^T \phi_n^{(c)}|^2$. At $B_c = 3$ ($t_c/g = 0.1$) the natural orbitals of ${}^c\Delta$ have occupations $\sim 10^{-7}$, 0.9999 and 1 (the nearly full core orbital and two pure ghosts), and R_c lies entirely along the core orbital (components along the others $\leq 10^{-4}$), so the occupied ghost level carries 3×10^{-7} and 4×10^{-10} at $V/g = 0.707$, 1.414 ($\leq 2.3\times10^{-9}$ at $B_c = 5$), and $R_c^T R_c = 0.994$, 0.977 sits on one pole. Mixing the two filled core orbitals by hand in λ_c does split the weight (e.g. 0.57/0.42 at $V/g = 0.707$, $\delta = 1$), but $\|[H_{qp},\rho]\|$ then grows from $\leq 2.6\times10^{-5}$ to 10^{-2}–4×10^{-2}: these points are not stationary, and their poles and weights do not match the exact ones. The ghosts appear to carry the correlation through ν_c and K (the terms $-V(\bar\Theta u)R_o^T$, $-V(\bar\Theta w)R_v^T$ of H_{qp}) but not through R_c, which is all that Eq. (19) sees.

Muted poles. In gGut every pole of the spectral function is an eigenvalue of H_{qp}, which has $B_c + B_o + B_v$ levels. Three of them are the quasiparticle (QP) levels of c, o and v, the levels where the orbital c, o or v has its largest weight: the core hole near -10, the occupied level near $-g$ and

the virtual level near $+g$. The remaining $B_c+B_o+B_v-3$ levels are the *muted poles*: two at $B_c=3$ (with $B_o=B_v=1$) and four at $B_c=5$. Occupied ones are hole poles, empty ones particle poles. Their weight $|R_I^T\phi_n^{(I)}|^2$ is $\leq 2.2\times10^{-6}$ on every orbital, so they are invisible in $A(\omega)$. They matter because they are the only place where gGut could put a satellite: a satellite needs a pole of H_{qp} at $\omega\simeq-12$ or lower, with weight.

- *Character.* The hole-side ones are ghost levels of the core block (>0.99), with core weight $<10^{-4}$. They are counted by occupation, not by energy order, since a free ghost level can cross the level o (in the $B_c=5$ solution at $t_c/g=0.05$, $V/g=1.414$ the o level is empty and a ghost level is occupied). The Fig. 2 dots are the occupied ones: one at $B_c=3$, two at $B_c=5$. The empty level at $\omega\simeq+12$ (Fig. 3) is not a ghost: its weight sits on o ($\simeq10^{-7}$–10^{-6}) and it matches the exact o particle peak at 12.003.

- *Position.* It is not fixed by the solution: the constraint Jacobian has rank 7 of 8 ($B_c=3$) and 13 of 17 ($B_c=5$), and shifting the multipliers along the undetermined directions moves the ghost levels, for example to the exact satellite energies, without changing E, the main pole, its weight or $\|[H_{qp},\rho]\|$; their weights stay $\leq 8\times10^{-9}$. Different minimizer solutions of the same point place them differently (-2.98 and -5.93 in the figures, -2.20 and -9.53 in earlier solutions, at $B_c=3$, $t_c/g=0.05$); the returned values are the minimum-norm multipliers. They lie between the core level (≈-10) and the valence region (≈-1), because the ghost levels come from the multipliers of the density constraint, which are of valence-scale magnitude (the bath levels of λ_c^c at $t_c/g=0.5$ are -2.1 to $+1.7$ and -5.1 to $+3.0$; `excited_phi_cederbaum.tex`), and not from E_c.

- *Weight.* It is set by R_c, which the energy minimization fixes along the core orbital. A solution with weight on a muted pole would have a different R_c and so differ from the minimum; none was found.

- *Ghosts on o and v.* With $B_c=B_o=B_v=3$ (9 levels, 6 muted poles) I seeded the minimization from the $B_c=3$ solution plus a sleeping ghost pair on o and on v (same energy), perturbed it (15 runs per case), and also tried random starts. No run went below the seed energy; the random starts mostly fell to the trivial product state ($R_c=0$). In three of the four cases of Fig. 3 a run returns to the $B_c=3$ point ($E-E_{\rm ED}\simeq9.5\times10^{-8}$, same main pole and weight); at $V/g=0.707$, $t_c/g=0.1$ the best point found is $B_c=1$-like (6.4×10^{-6}). The extra poles again carry no weight (purple row of Fig. 3). The search is limited (SLSQP, seeds from $B_c=3$ only).

- **Note on the search for an entangled Φ_c.** A satellite needs R_c on several qp levels, i.e. ${}^c\Delta$ with partially occupied ghost natural orbitals (here 0, 0.99998, 1). I seeded Φ_c with the first four eigenstates of random embedding Hamiltonians with strong core–ghost hybridization (scales 0.5, 1, 2; 3 seeds each; Φ_o, Φ_v, X from the $B_c=3$ solution; 36 runs per V/g at $t_c/g=0.05$; `search_entangled_phi_c.py`). Converged runs (15 at $V/g=0.707$, 4 at 1.414) all return to the known point ($E-E_{\rm ED}\simeq9.5\times10^{-8}$, one pole of weight 0.994 and 0.977); the others stop away from a stationary point. No entangled solution with weight off the main pole was found; constraining that weight directly (`test_pole_weight_constraint.py`) did not reach it either. This does not exclude one: only 72 seeds, with o and v fixed.

The exact satellites lie *below* the main peak (-12 and -14.5 at $V/g=0.707$, -15.5 at $V/g=1.414$, against -9.6 exact, -9.9 gGut), whereas the muted poles lie above it, so none of them can play the role of a satellite.

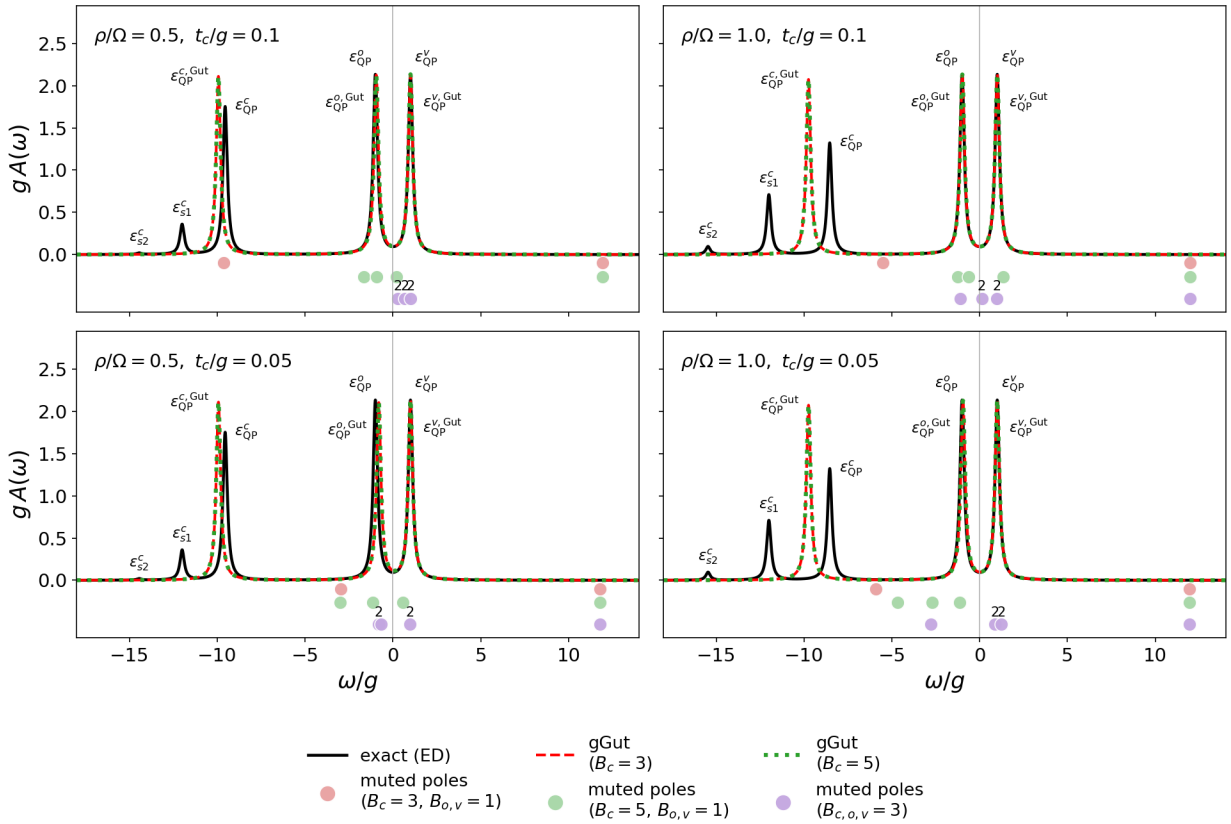


Figure 3: Full spectral function $A(\omega) = \sum_{I=c,o,v} A_{II}(\omega)$ (hole side $\omega < 0$, particle side $\omega > 0$; per spin, $\eta = 0.15$, $E_c/g = 10$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$, 1.0. *Curves:* exact (black), gGut $B_c = 3$ (red dashed) and $B_c = 5$ (green dotted; both with $B_o = B_v = 1$, from the cached solutions of Fig. 2). *Labels:* $\varepsilon^I_{\mathrm{QP}}$ is the quasiparticle peak of the exact spectrum for the orbital $I = c, o, v$ and ε^c_{s1}, ε^c_{s2} are the exact core satellites, numbered by increasing distance from $\varepsilon^c_{\mathrm{QP}}$; $\varepsilon^{I,\mathrm{Gut}}_{\mathrm{QP}}$ are the gGut quasiparticle peaks (the core one is single, with no satellites). *Dots below the axis:* the muted poles of gGut, i.e. all eigenvalues of H_{qp} except the three quasiparticle levels. One row per solution: $B_c = 3$ (rose, 2 poles), $B_c = 5$ (green, 4) and $B_c = B_o = B_v = 3$ (purple, 6; the lowest-energy run seeded from the $B_c = 3$ solution with sleeping ghosts on o and v; in the panel $\rho/\Omega = 0.5$, $t_c/g = 0.1$ no run returned to the seed and the best point is $B_c = 1$-like). Poles closer than 0.02 share one dot, with the multiplicity written above it when it is larger than 1. The muted poles carry no weight ($\leq 2.2 \times 10^{-6}$) and none lies near the exact satellites.

Full spectral function (core, occupied, virtual). Fig. 3 shows all peaks of $A = A_{cc} + A_{oo} + A_{vv}$, hole side ($\omega < 0$) and particle side ($\omega > 0$), against ED. In gGut every eigenvalue of H_{qp} is a pole, occupied ones on the hole side and empty ones on the particle side, with weight $|R_I^T \phi_n^{(I)}|^2$ on orbital I. The three main peaks are the core hole near -10 (c), the occupied level near $-g$ (o) and the virtual level near $+g$ (v); gGut reproduces the last two, and the core pole only approximately (position and weight as in Tab. 3). What gGut misses are the exact core satellites at -12 and -14.5 (-15.5); the remaining ED peaks ($\omega \simeq 3$ for c and $\omega \simeq 12$ for o, particle side) have weight $\lesssim 10^{-6}$ and are not visible. The remaining levels are the muted poles defined above ($B_c = 3$: one hole pole at -5.9 to -3.0 and the empty level at $\simeq +12$; $B_c = 5$: several between -4.7 and $+1.4$ besides that one), all with weight $\leq 10^{-6}$.

6.3 Comparison with the three-level model of Cederbaum and of Kockläuner and Golze

The orbitals c, o, v are the three-level core-hole model of Cederbaum, Domcke and Schirmer [1] (Eqs. (28)–(29)), used by Kockläuner and Golze [2] (Fig. 2) to test G_0W_0 and GW plus cumulant (GW+C). With $\Omega = \epsilon_v - \epsilon_o = 2g$ and the transition moment ρ (GW+C excitation weight $\gamma = \rho^2/\Omega^2$), the exact solution is the 3×3 matrix with off-diagonal elements $-\sqrt{2}V$, which gives Eq. (3) and the weights $(a \pm \Omega)^2/4a^2$, $2V^2/a^2$ of Ref. [1] ($a = (\Omega^2 + 4V^2)^{1/2}$). The $\sqrt{2}$ is the normalization of the two spin channels, so

$$\rho = \sqrt{2}\, V, \qquad \frac{\rho}{\Omega} = \frac{V}{\sqrt{2}\, g}, \tag{20}$$

confirmed by the small-V limit of the exact satellite-to-QP weight ratio ($2.00\times 10^{-4} = (\sqrt{2}V)^2/\Omega^2$ at $V/g = 0.02$, `g0w0_cederbaum.py`). Their ratios $\rho/\Omega = 0.5$, 1.0 are $V/g = 0.707$, 1.414, the values used throughout this note. Their figure gives only the ratio, no energy axis, so E_c cannot be taken from it (it only shifts the poles rigidly); Fig. 4 therefore uses ω with ticks at the exact ϵ_c^{QP}, $\epsilon_c^{\mathrm{Sat1}}$, $\epsilon_c^{\mathrm{Sat2}}$.

G_0W_0 and GW+C for the model (computed here). We computed G_0W_0 and GW+C for this model ourselves (`g0w0_cederbaum.py`) from the expressions below, in order to compare with the G_0W_0 and GW+C of Ref. [2] (not taken from it). With G_0 the bare core level and one bosonic mode, $\Sigma_c(\omega) = \rho^2/(\omega - \epsilon_c + \Omega)$ (no exchange or Hedin shift), the poles of $[\omega - \epsilon_c - \Sigma_c]^{-1}$ solve $x(x+\Omega) = \rho^2$: a QP pole and one satellite at $-\Omega\sqrt{1 + 4\rho^2/\Omega^2}$, against the exact satellites at $-\Omega\sqrt{1 + 2\rho^2/\Omega^2}$ and twice that. GW+C has Poisson satellites at $\epsilon_{\mathrm{QP}} - n\Omega$ with weights $e^{-\gamma}\gamma^n/n!$. These expressions agree with Eqs. (11), (14), (16)–(17), (33), (36), (44)–(47) of Ref. [2] (the exchange constant $\Sigma^x - v^{xc}$ is absorbed in ϵ_c), and the first-satellite positions measured in their rendered Fig. 2b,c (in units of Ω, accuracy a few per cent) agree with this note: exact 1.22 and 1.76 (here 1.225, 1.732), G_0W_0 1.42 and 2.23 (here 1.414, 2.236) at $\rho/\Omega = 0.5$, 1.0.

ρ/Ω		satellite positions /Ω (from QP)	weights (QP, satellites)
0.5	exact	-1.225, -2.449	0.825, 0.167, 0.008
	G_0W_0	-1.414	0.854, 0.146
	GW+C	-1, -2, -3, …	0.779, 0.195, 0.024, …
1.0	exact	-1.732, -3.464	0.622, 0.333, 0.045
	G_0W_0	-2.236	0.724, 0.276
	GW+C	-1, -2, -3, …	0.368, 0.368, 0.184, …

Table 4: Exact ($t_c = 0$), G_0W_0 and GW+C hole spectrum of the model at the ratios of Fig. 2 of Ref. [2] ($V/g = 0.707$, 1.414).

What matches. The exact spectrum has the QP peak and two satellites (a single and a double $o \to v$ excitation, the second much weaker), with intensity and QP–satellite separation growing with ρ/Ω; G_0W_0 has one satellite at higher binding energy than the exact one ($-1.41\,\Omega$ against $-1.23\,\Omega$ at $\rho/\Omega = 0.5$; $-2.24\,\Omega$ against $-1.73\,\Omega$ at 1.0); GW+C puts satellites at $-n\Omega$ (closer to the exact ones at $\rho/\Omega = 0.5$ than at 1.0) and shows spurious higher satellites at larger ρ/Ω; all as stated in Ref. [2]. The hopping t_c moves the exact poles by at most 0.06 at $t_c/g = 0.5$.

What does not match. gGut gives a single pole at both ratios (weight 0.977–0.9999): it reproduces neither the exact satellites nor the GW+C or G_0W_0 ones, and does worse than G_0W_0 for

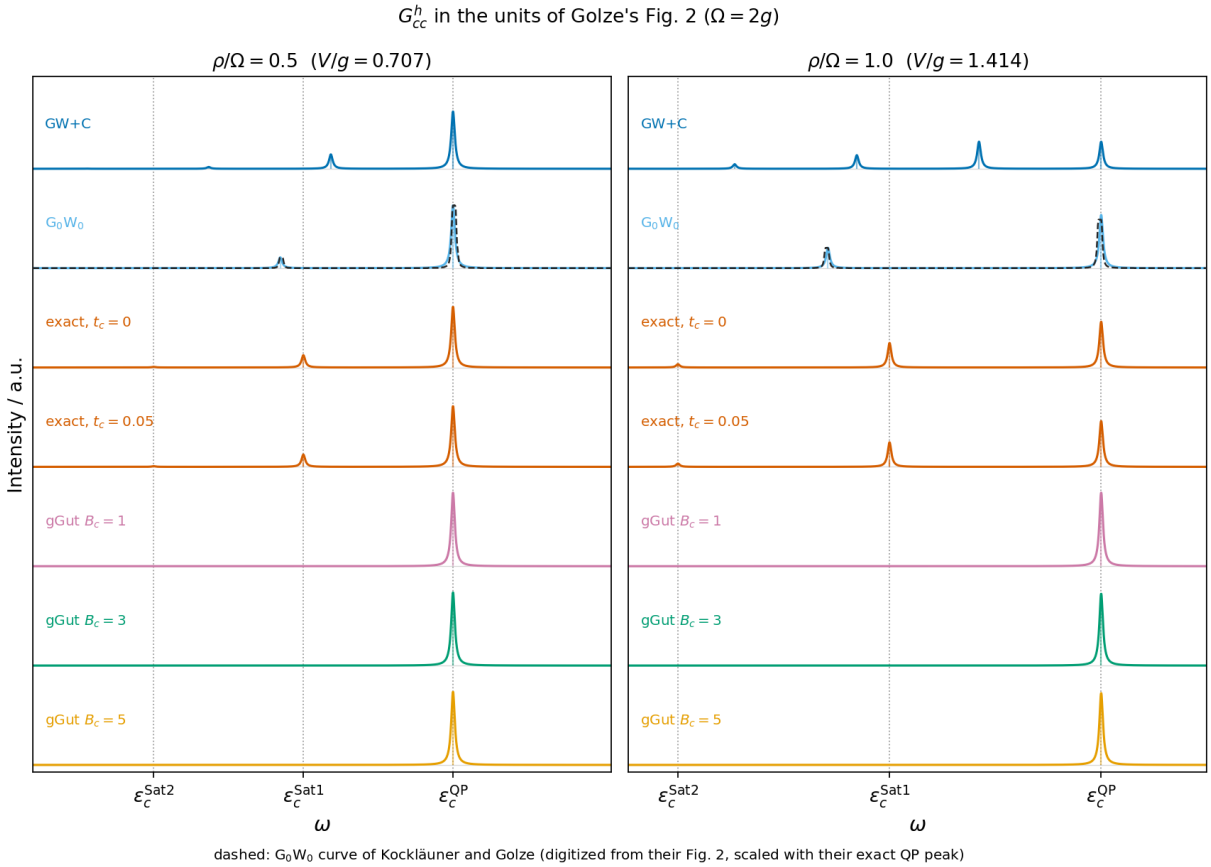


Figure 4: G^h_{cc} in the units of Fig. 2 of Ref. [2]; left $\rho/\Omega = 0.5$ ($V/g = 0.707$), right $\rho/\Omega = 1.0$ ($V/g = 1.414$), $E_c/g = 10$. Rows: GW+C; G_0W_0; exact at $t_c = 0$ and 0.05; gGut $B_c = 1$, 3, 5 ($t_c/g = 0.05$). All rows share one scale (peak height = pole weight) and a Lorentzian half-width $0.018\,\Omega$, fitted to the G_0W_0 peaks of Golze. Ticks: exact $t_c = 0$ QP and satellite energies (dotted lines). Dashed: the G_0W_0 curve of Ref. [2], digitized from their Fig. 2 and scaled with their exact QP peak (`measure_golze_fig2.py`); its QP peak at $\rho/\Omega = 1.0$ is cut off by their frame.

this model, whose QP pole is within 0.04 and 0.23 of the exact one (-9.59, -8.76 against -9.55, -8.54) while the gGut $B_c = 3$ pole is at -9.93, -9.74 (errors 0.38, 1.21) and the $B_c = 1$ pole at -10.00. The muted poles (dots of Fig. 2) are not satellites: no weight and arbitrary positions, as discussed above.

7 Conclusions

1. **gGut does not reproduce the satellite structure of the model.** For $B_c = 1$, 3, 5, $\rho/\Omega = 0.5$, 1.0 ($V/g = 0.707$, 1.414) and $t_c/g = 0.05$–0.5, the hole spectral function has a single pole with weight 0.977–0.9999, whereas the exact spectrum has satellites near -12 and at -14.45, -15.47 carrying 17.5% and 37.8% of the weight. The main pole is displaced from the exact one (-9.93 vs. -9.55; -9.74 vs. -8.54).

2. The ghosts improve the ground state: $E - E_{\mathrm{ED}} \approx 10^{-7}$ at $B_c = 3$ and 5 (vs. $1.3 \times 10^{-3} V^2 t_c^2$ at $B_c = 1$), ν_c to $\sim 10^{-8}$; $B_c = 5$ returns the $B_c = 3$ energy and spectrum (the extra ghosts

stay asleep).

3. The muted poles (the levels of H_{qp} beyond the three QP levels) have zero weight (R_c lies along the core orbital) and arbitrary energy (free multiplier directions): moving them to the exact satellite energies leaves E and the weights unchanged, so they give no satellite. A solution with satellite weight would have to change R_c, i.e. differ from the energy minimum; none was found.

4. The model uses a core–virtual hopping: a core–occupied hopping leaves $|220\rangle$ exact for every t_c and cannot make R_c determined. The energy functional and its gradients are verified (finite differences $\sim 10^{-10}$), and the limits $V = 0$, $t_c = 0$ are reproduced.

5. The model is the three-level model of Cederbaum *et al.* [1] with $\rho = \sqrt{2}V$; for $\rho/\Omega = 0.5$, 1.0 the exact spectrum has the QP peak and two satellites, G_0W_0 one satellite at too high a binding energy, GW+C satellites at $-n\Omega$, and gGut a single pole (Fig. 4, Tab. 4).

A The quasiparticle Hamiltonian as a matrix

H_{qp} acts on the B_c+2 quasiparticle orbitals of one spin, ordered as the B_c core orbitals $d_{c1}, \ldots, d_{cB_c}$, then d_o, then d_v (the block structure c, o, v of ρ in Sec. 3). It is real and symmetric. Its ingredients, all defined in Sec. 3, are the renormalization vectors $R_c \in \mathbb{R}^{B_c}$, $R_o, R_v \in \mathbb{R}$, the core occupation ν_c, the tensor $\bar{\Theta} \in \mathbb{R}^{B_c \times B_c}$ and the two core vectors

$$a = \bar{\Theta}\, u = \bar{\Theta}\, \rho^{cv} R_v, \qquad b = \bar{\Theta}\, w = \bar{\Theta}\, \rho^{co} R_o \qquad (a, b \in \mathbb{R}^{B_c}). \tag{21}$$

The diagonal blocks are the multipliers of the constraint $\rho^{II} = {}^{I}\Delta$ of Eq. (9), $-\lambda_c \in \mathbb{R}^{B_c \times B_c}$ (symmetric) and $-\lambda_o, -\lambda_v \in \mathbb{R}$; the off-diagonal blocks are $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$ of Sec. 4. In block form, with the blocks of size B_c, 1, 1,

$$H_{qp} = \begin{pmatrix} -\lambda_c & -V R_o\, a & R_v\,(t_c R_c - V\, b) \\ -V R_o\, a^T & -\lambda_o & -V(2-\nu_c)\, R_o R_v \\ R_v\,(t_c R_c - V\, b)^T & -V(2-\nu_c)\, R_o R_v & -\lambda_v \end{pmatrix}. \tag{22}$$

The *co* and *cv* blocks are the column vectors $-V R_o\, a$ and $R_v(t_c R_c - V b)$ ($B_c \times 1$); the *ov* block is a number. This is what `build_Hqp` returns (`gga_cederbaum.py`), and it is the same operator as $H_{\text{off}} - \text{blockdiag}(\lambda_I)$ of Sec. 4.

Spin. H_{qp} and ρ in this appendix are the *one-spin* matrices, of size $B_c + 2$: $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle$ with σ fixed, and the solutions are paramagnetic, so $\uparrow$ and $\downarrow$ have the same matrix. The quasiparticle Hamiltonian on the full space of $2(B_c + 2)$ spin orbitals, ordered ($\uparrow$ block, $\downarrow$ block), is

$$H^{\text{full}}_{qp} = \begin{pmatrix} H_{qp} & 0 \\ 0 & H_{qp} \end{pmatrix} = \mathbb{1}_2 \otimes H_{qp}, \qquad \rho^{\text{full}} = \mathbb{1}_2 \otimes \rho, \tag{23}$$

and each spin channel fills its N_{occ} lowest levels ($N_{\text{occ}} = 2$ for $B_c = 1$, 3 for $B_c = 3$), i.e. $2N_{\text{occ}}$ electrons in total. The spin sum and the hermitian conjugate that give the factor 4 in Eq. (7) are removed in $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$, so the entries below carry no spin factor.

Explicit form for $B_c = 1$. There are no ghosts, λ_c, a, b, R_c, $\bar{\Theta}$ are numbers and H_{qp} is 3×3:

$$H_{qp}^{B_c=1} = \begin{pmatrix} -\lambda_c & -VR_o\, a & R_v(t_c R_c - V\, b) \\ -VR_o\, a & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v(t_c R_c - V\, b) & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}, \tag{24}$$

with $a = \bar{\Theta} R_v \rho^{cv}$ and $b = \bar{\Theta} R_o \rho^{co}$.

Diagonalizing it, $H_{qp}\phi_n = \epsilon_n \phi_n$ ($\epsilon_1 \le \epsilon_2 \le \epsilon_3$, $\phi_{n,p}$ the component of ϕ_n on orbital p), and filling the $N_{\mathrm{occ}} = 2$ lowest levels per spin gives $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle = \sum_{n=1}^{2} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=1} = \begin{pmatrix} s_{cc} & s_{co} & s_{cv} \\ s_{co} & s_{oo} & s_{ov} \\ s_{cv} & s_{ov} & s_{vv} \end{pmatrix}. \tag{25}$$

The diagonal entries are the new ${}^I\Delta$, and the off-diagonal ones give the a, b of the next H_{qp} through $u = R_v s_{cv}$, $w = R_o s_{co}$: $a = \bar{\Theta} R_v s_{cv}$, $b = \bar{\Theta} R_o s_{co}$.

Explicit form for $B_c = 3$. With $\lambda_c = (\lambda_{ij})$ symmetric, $a = (a_1, a_2, a_3)^T$, $b = (b_1, b_2, b_3)^T$, $R_c = (R_{c1}, R_{c2}, R_{c3})^T$ and $\tau_i = t_c R_{ci} - V b_i$, H_{qp} is 5×5:

$$H_{qp}^{B_c=3} = \begin{pmatrix} -\lambda_{11} & -\lambda_{12} & -\lambda_{13} & -VR_o a_1 & R_v\tau_1 \\ -\lambda_{12} & -\lambda_{22} & -\lambda_{23} & -VR_o a_2 & R_v\tau_2 \\ -\lambda_{13} & -\lambda_{23} & -\lambda_{33} & -VR_o a_3 & R_v\tau_3 \\ -VR_o a_1 & -VR_o a_2 & -VR_o a_3 & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v\tau_1 & R_v\tau_2 & R_v\tau_3 & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}. \tag{26}$$

Diagonalizing it ($\epsilon_1 \le \cdots \le \epsilon_5$) and filling the $N_{\mathrm{occ}} = 3$ lowest levels per spin gives $\rho_{pq} = \sum_{n=1}^{3} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=3} = \begin{pmatrix} s_{c_1c_1} & s_{c_1c_2} & s_{c_1c_3} & s_{c_1o} & s_{c_1v} \\ s_{c_1c_2} & s_{c_2c_2} & s_{c_2c_3} & s_{c_2o} & s_{c_2v} \\ s_{c_1c_3} & s_{c_2c_3} & s_{c_3c_3} & s_{c_3o} & s_{c_3v} \\ s_{c_1o} & s_{c_2o} & s_{c_3o} & s_{oo} & s_{ov} \\ s_{c_1v} & s_{c_2v} & s_{c_3v} & s_{ov} & s_{vv} \end{pmatrix}. \tag{27}$$

The upper-left 3×3 block is the new ${}^c\Delta$ and s_{oo}, s_{vv} the new ${}^o\Delta$, ${}^v\Delta$. The off-diagonal entries give the a, b of the next H_{qp}: $u = R_v(s_{c_1v}, s_{c_2v}, s_{c_3v})^T$, $w = R_o(s_{c_1o}, s_{c_2o}, s_{c_3o})^T$, $a = \bar{\Theta} u$, $b = \bar{\Theta} w$.

For $B_c = 5$ the core block is 5×5 and the two border columns have five entries each, same pattern.

Use in the impurity loop. In the impurity loop (`gga_cederbaum.py`) H_{qp} is diagonalized once per macroiteration; there is no inner loop on ρ. The ρ obtained from the occupied eigenvectors is used directly in the next macroiteration: its diagonal blocks ρ^{II} are the ${}^I\Delta$ that enter the embedding problems (before the linear mixing with their output), and its off-diagonal blocks give u, w and hence a, b in the next H_{qp}. ρ is thus self-consistent with H_{qp} only at convergence of the macroiterations. (Repeating the diagonalization until ρ stops changing, before the embedding step, was tested and does not change the converged result for $B_c = 1$ nor make the $B_c = 3$ loop converge.)

In the direct minimization (Sec. 4) no such cycle is needed: $\rho = X(X^TX)^{-1}X^T$ is a variable of the minimization, E already contains the dependence of H_{qp} on ρ through u and w, and the

condition that ρ is built from its own H_{qp} is the stationarity condition $[H_{qp}, \rho] = 0$; the link $\rho^{II} = {}^{I}\Delta(\Phi_I)$ is an equality constraint. At the minimizers found, building H_{qp} from the multipliers and from this ρ, diagonalizing it and filling the N_{occ} lowest levels returns the same ρ to the accuracy of the stationarity ($\|\rho_{\mathrm{filled}} - \rho\| = 10^{-8}$ for $B_c = 1$ and for $B_c = 3$ at $\varepsilon = 10^{-5}$; 2×10^{-5} for $B_c = 3$ at $\varepsilon = 10^{-9}$, where $\|[H_{qp}, \rho]\| = 3 \times 10^{-4}$), and the occupied levels are the lowest ones, with a gap 1–2 to the first empty level ($V/g = 1$, $t_c/g = 0.5$).

References

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